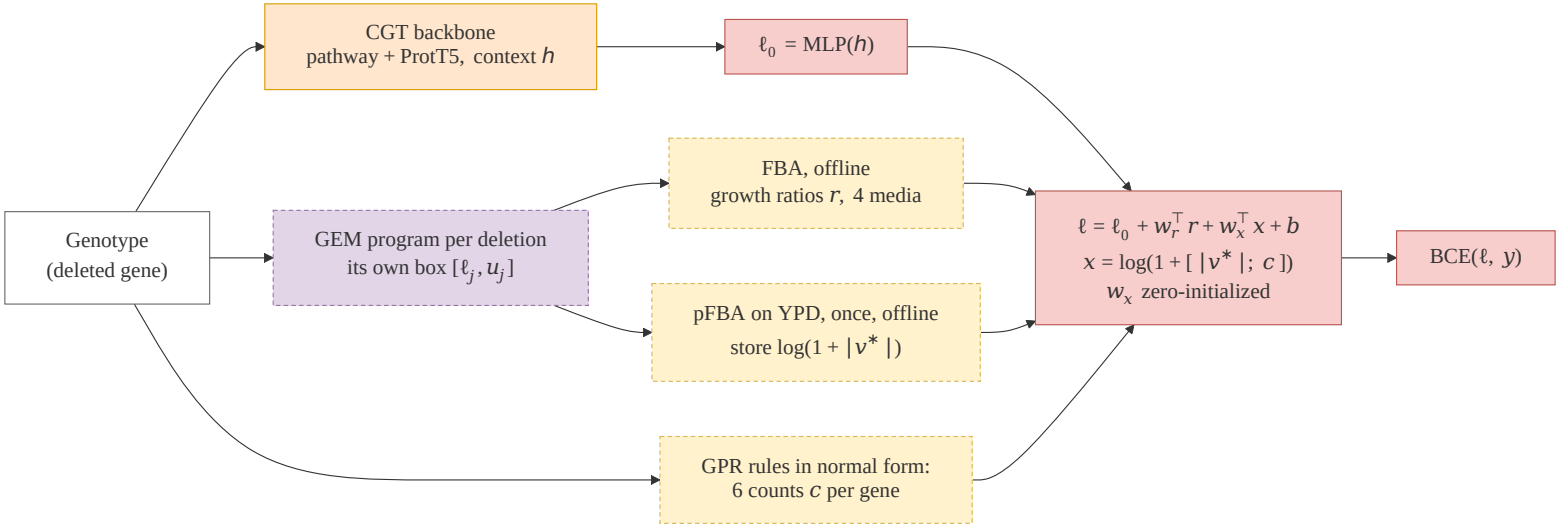

CGT backbone
pathway + ProtT5, context h
ℓ0 = MLP(h)
Genotype
(deleted gene)
GEM program per deletion
its own box [ℓj, uj]
FBA, offline
growth ratios r, 4 media
pFBA on YPD, once, offline
store log(1 + |v*|)
GPR rules in normal form:
6 counts c per gene
ℓ = ℓ0 + wr⊤ r + wx⊤ x + b
x = log(1 + [ |v*| ; c ])
wx zero-initialized
BCE(ℓ, y)